AI × FINANCE × DIGITAL TWIN

FinTwin AI

AI-Powered Financial Health Digital Twin with Scenario Simulation and Behaviour Coaching

PROJECT REVIEW — 02

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M.Sc. Data Science — Semester 3

Project Overview

02

01 AI-powered personal financial decision-support system

02 Creates a Financial Digital Twin of the user

03 Analyzes income, expenses, savings, investments and debt

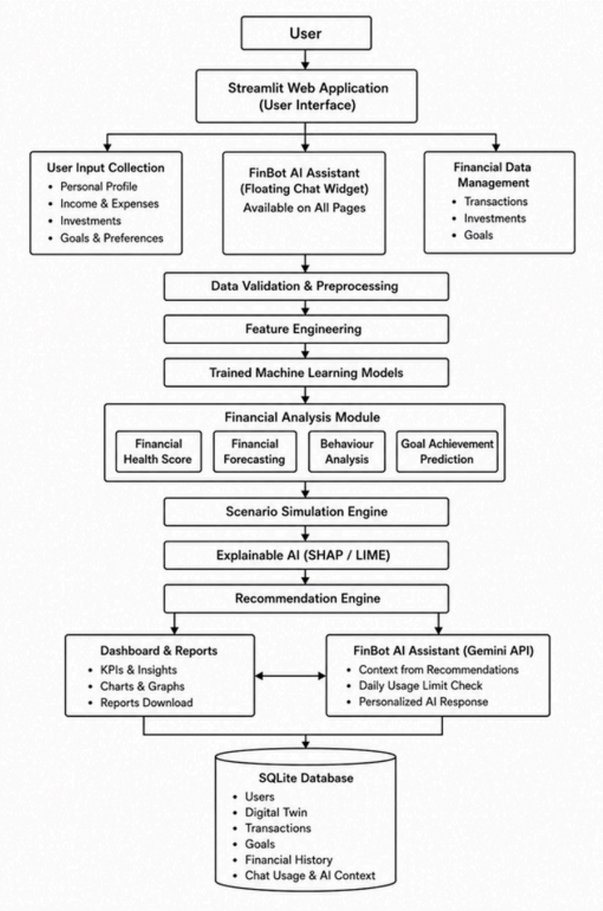
04 Provides predictive and personalized financial insights

05 Uses Machine Learning and Explainable AI

KEY FEATURES

- Financial Health
- Behaviour Analysis
- Forecasting
- Scenario Simulation
- AI Recommendations
- XAI

System Architecture



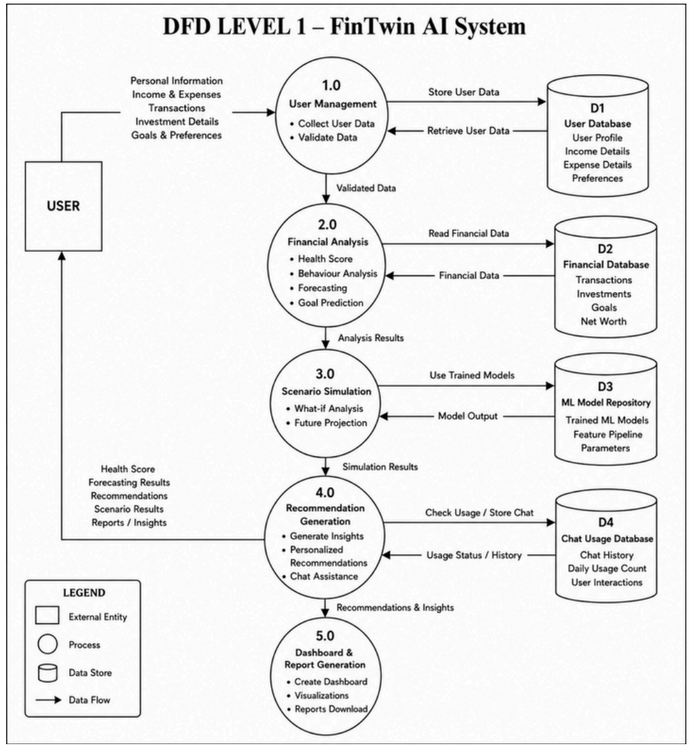
TECHNOLOGY STACK

- Python
- Streamlit
- Scikit-learn
- XGBoost
- Pandas
- Plotly
- SHAP

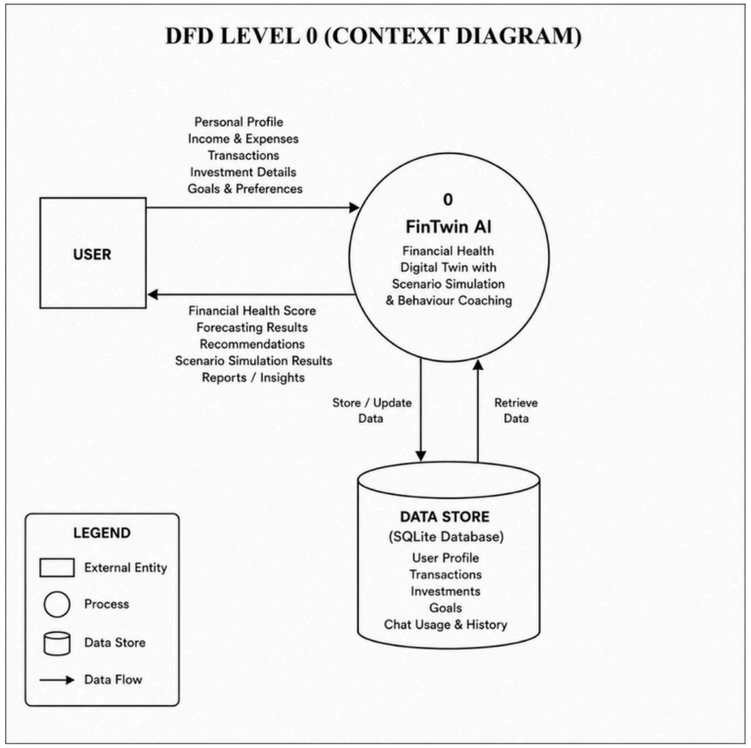
Data Flow Diagram

INPUT | PROCESSING | ANALYSIS | OUTPUT

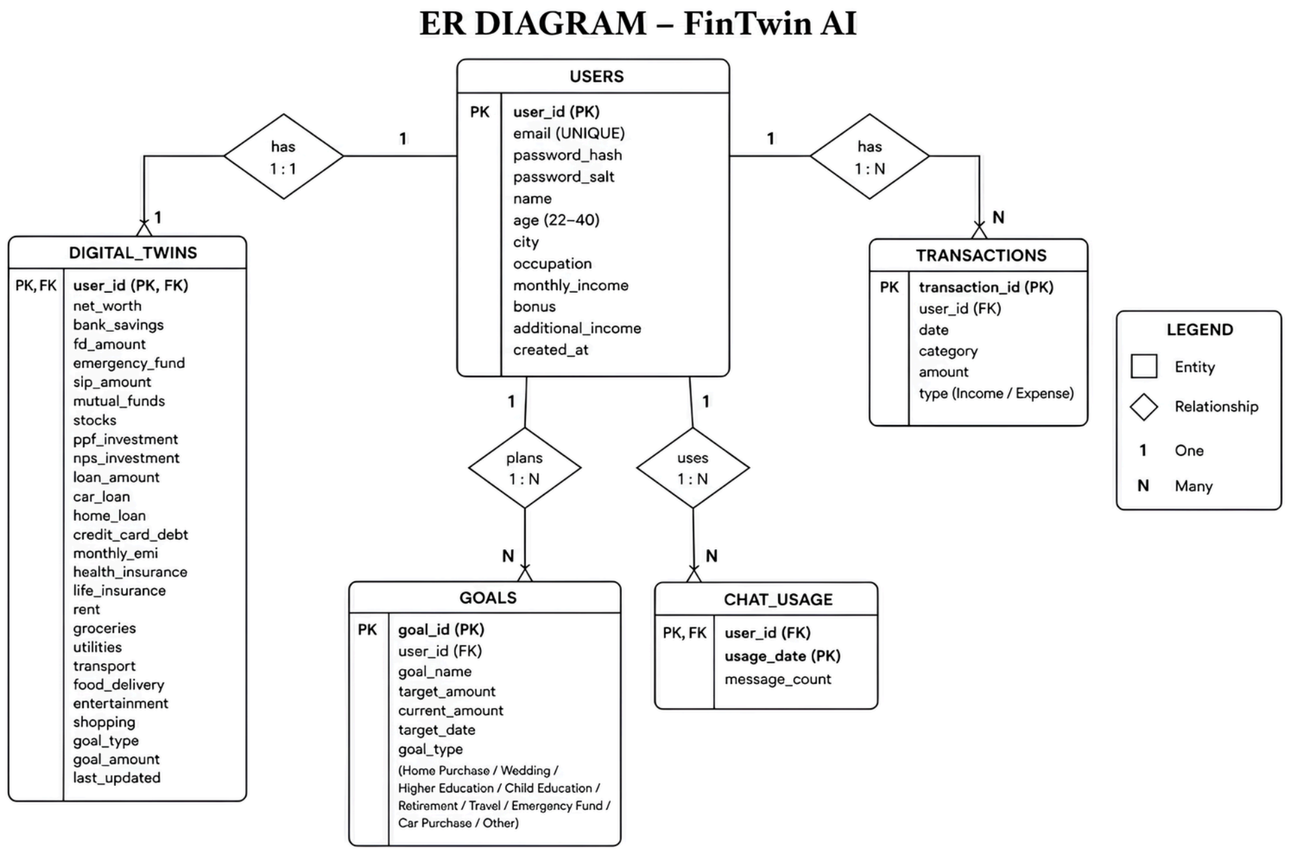
LEVEL 1 — DETAILED



LEVEL 0 — CONTEXT



ER Diagram & Database Design



MAIN ENTITIES

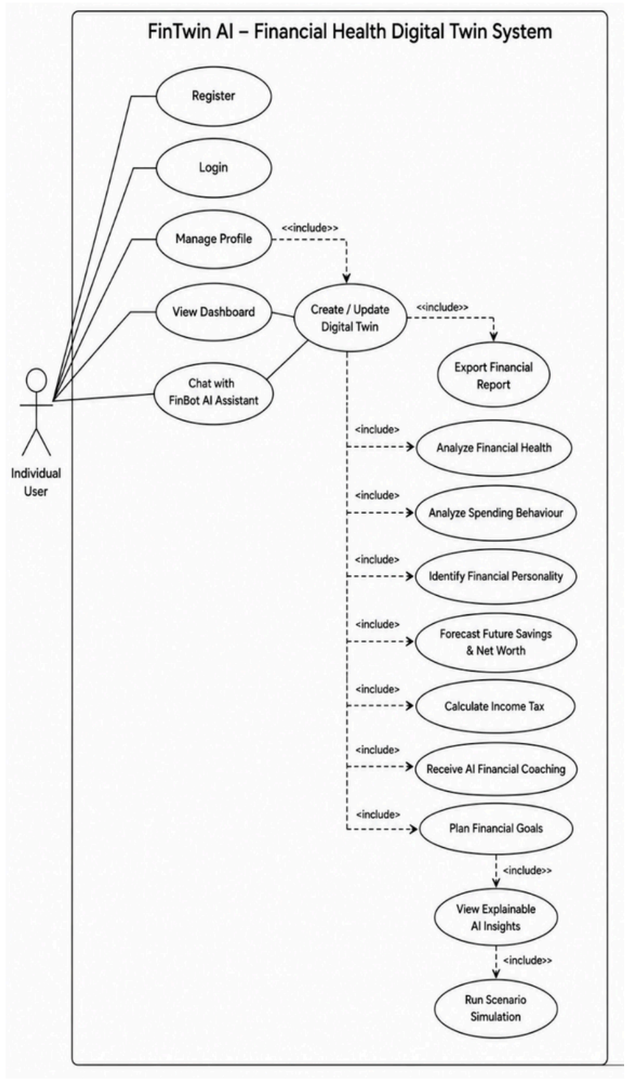
- Users
- Digital Twins
- Transactions
- Goals
- Chat Usage

DATABASE PURPOSE

Store and organize user financial information for analysis, prediction and personalized recommendations.

UML Design

USE CASE DIAGRAM

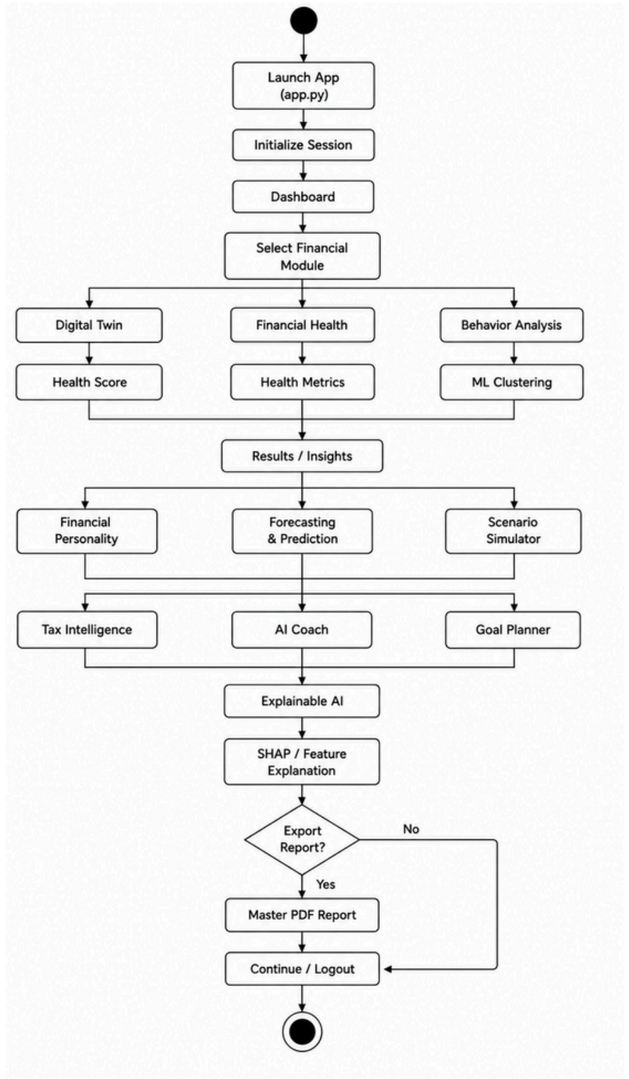


MAIN ENTITIES

- Individual User
- User Profile
- Financial Digital Twin
- AI/ML Analysis Engine
- Financial Insights & Recommendations

UML Design

ACTIVITY DIAGRAM

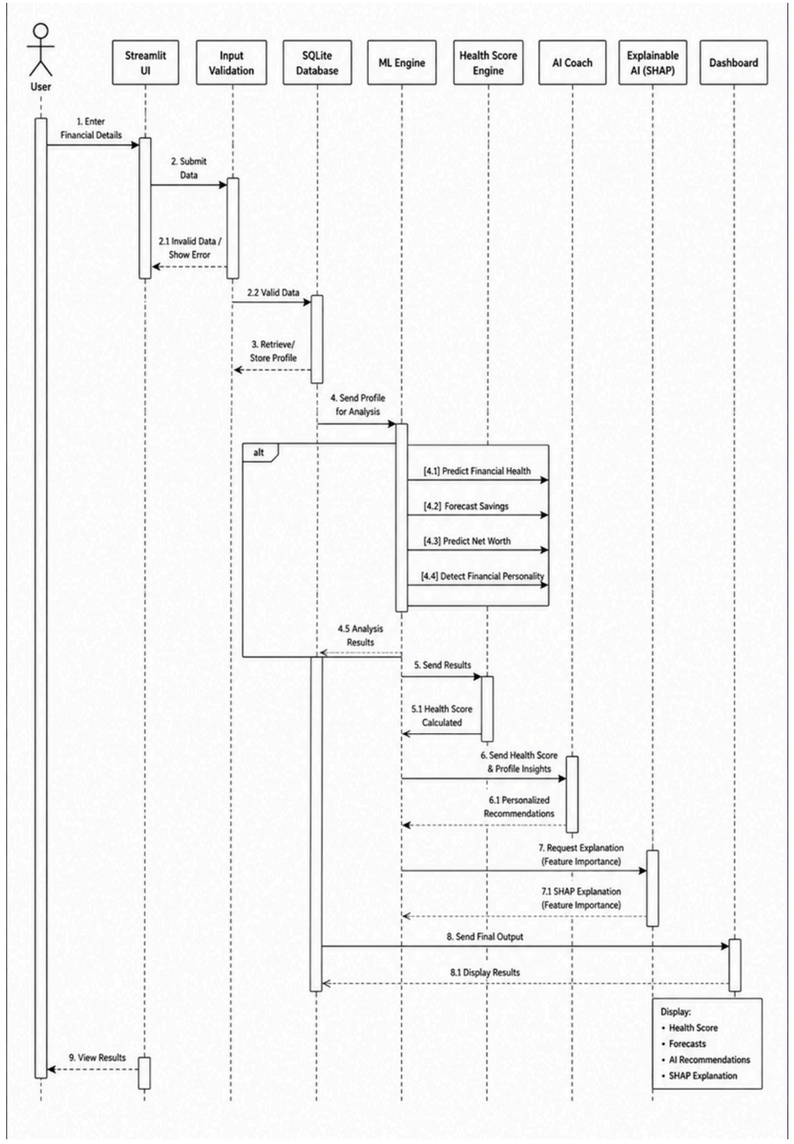


MAIN ENTITIES

- Application
- Financial Digital Twin
- Analysis Engine
- Forecasting & Simulation
- AI Coach & Goal Planner
- Explainable AI & Reporting

UML Design

SEQUENCE DIAGRAM

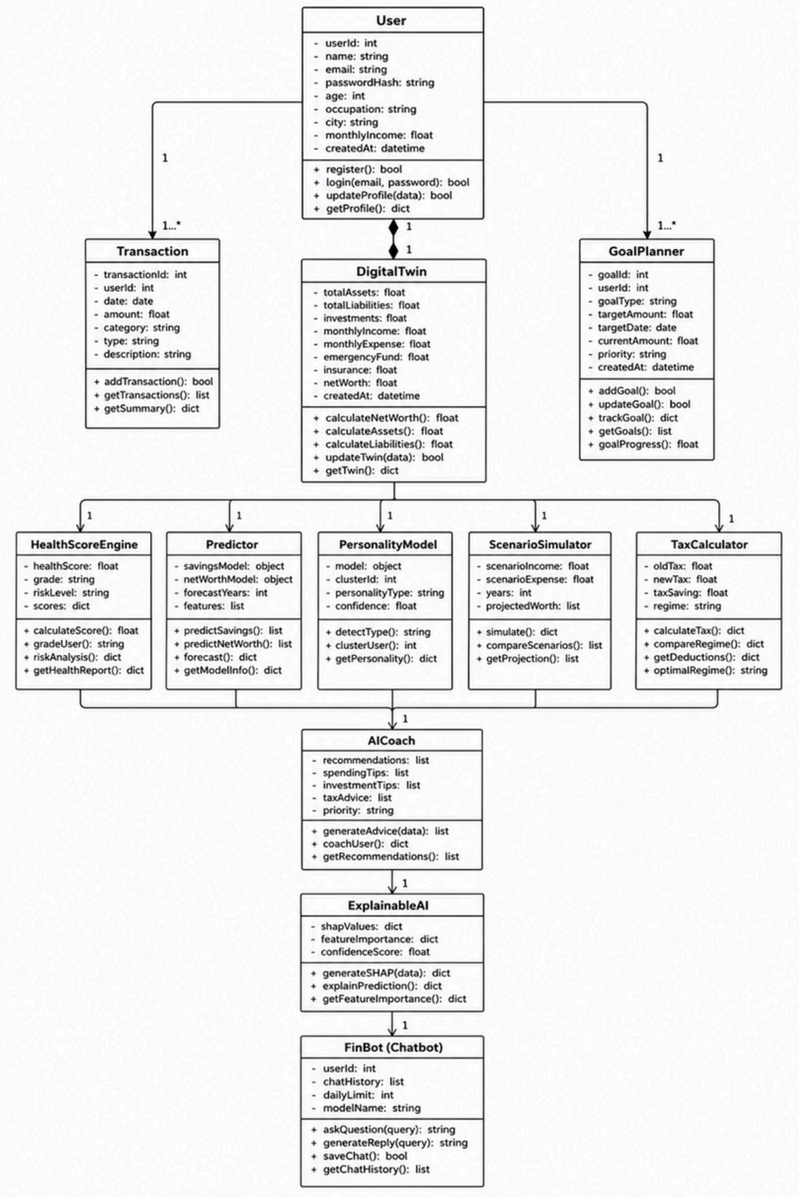


MAIN ENTITIES

- User
- Streamlit UI
- SQLite Database
- ML & Health Score Engine
- AI Coach + Explainable AI
- Dashboard

UML Design

CLASS DIAGRAM



MAIN ENTITIES

- User
- DigitalTwin
- ML Analysis Modules
- AI Coach
- Explainable AI
- FinBot

Module Planning

All modules are integrated into the FinTwin AI workflow.

Financial Profile

Collect financial information

Financial Health

Calculate health score

Behaviour Analysis

Analyze spending behaviour

Financial Personality

Identify financial pattern

Forecasting

Predict future outcomes

Goal Planner

Analyze financial goals

Scenario Simulator

Perform what-if analysis

AI Coach

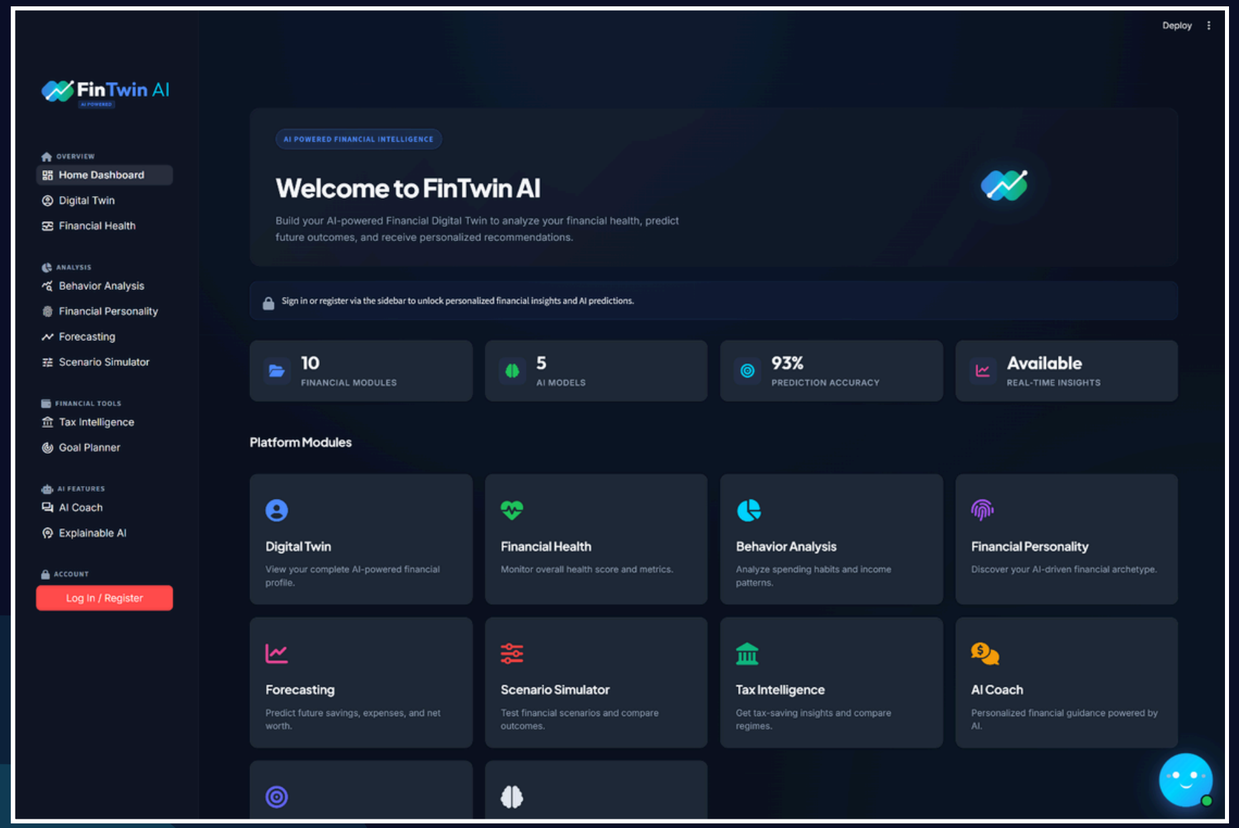
Provide personalized guidance

Explainable AI

Explain predictions

UI Prototype

HOME DASHBOARD



DIGITAL TWIN

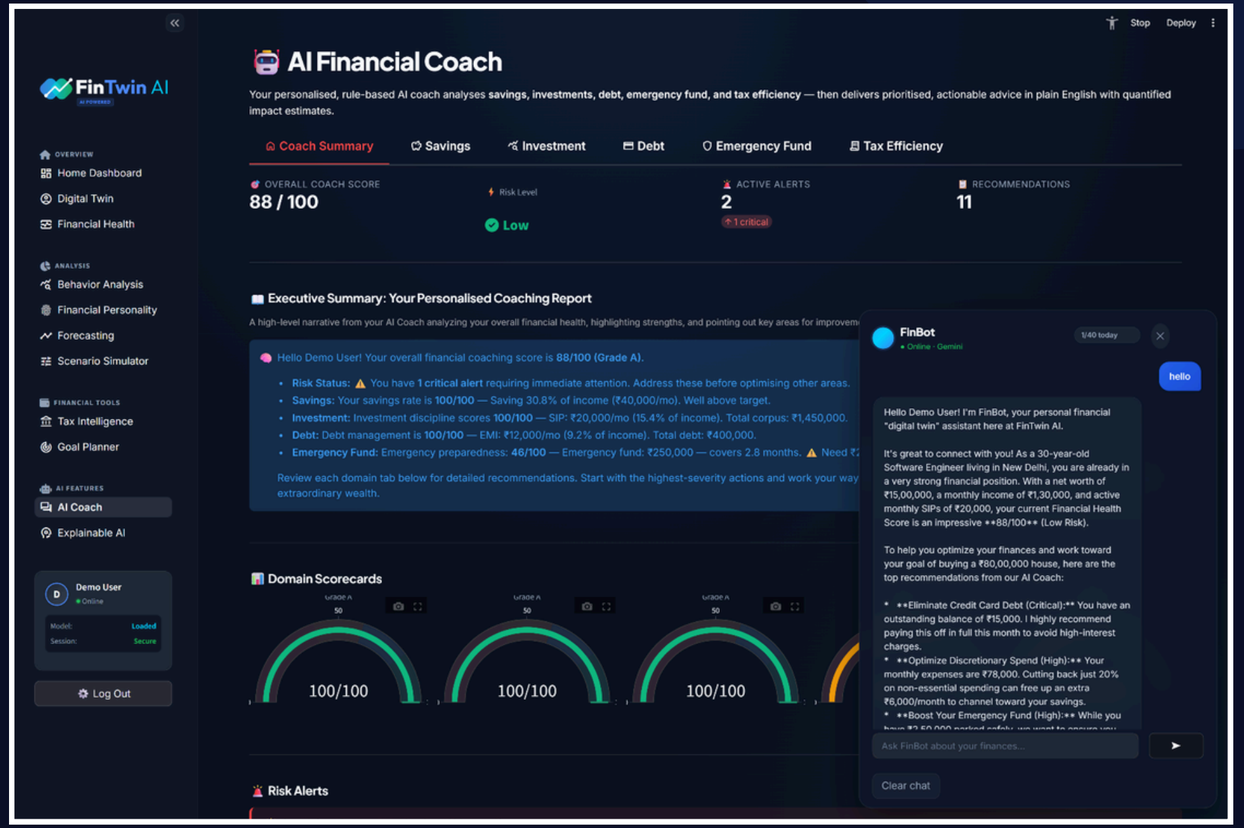


UI Prototype

FINANCIAL HEALTH



AI COACH



CURRENT STATUS

Review 2 Progress

DELIVERABLES COMPLETED

- ✓ System Architecture
- ✓ DFD
- ✓ ER Diagram
- ✓ Database Design
- ✓ UML Diagrams
- ✓ Module Planning
- ✓ UI Prototype

CURRENT PROJECT STATUS

Chapters 1–4

Completed

NEXT STAGE

Implementation & Testing





Thank You

FinTwin AI — Project Review 2